

API Reference & Endpoints

Base URL: /api/v1

Authentication: Bearer JWT & HttpOnly Cookies

Protocol: HTTPS JSON

1. Authentication & Identity Management (/api/v1/auth)

Method	Endpoint	Access	Description
POST	/api/v1/auth/signup	Public	Register new user account with bcrypt password hashing.
POST	/api/v1/auth/login	Public	Authenticate with email & password; returns JWT token, user object, and session cookie.
POST	/api/v1/auth/google	Public	Google OAuth 2.0 verification via ID token; auto-provisions student accounts or maps staff.
POST	/api/v1/auth/logout	Authenticated	Clear authenticated session cookies and invalidate client-side token.
GET	/api/v1/auth/me	Authenticated	Fetch currently authenticated user session details, role permissions, and profile metadata.
PUT	/api/v1/auth/profile	Authenticated	Update profile details like phone, department, bio, skills, and portfolio links.

2. Central Administration & Governance (/api/v1/admin)

Method	Endpoint	Access	Description
GET	/api/v1/admin/stats	Admin	Fetch institutional metrics including enrollments, attendance rates, and placement drives.
GET	/api/v1/admin/users	Admin	Retrieve all system users across all roles with status and department metadata.
PATCH	/api/v1/admin/users/:id/role	Admin	Dynamically update user role permissions.
DELETE	/api/v1/admin/users/:id	Admin	Permanently delete user account and profile records from MongoDB Atlas.
GET	/api/v1/admin/students	Admin	Retrieve enrolled student registry records with roll numbers, CGPA, and status.
POST	/api/v1/admin/students	Admin	Manually onboard a single student into the registry.
POST	/api/v1/admin/students/bulk-import	Admin	Bulk import students via Excel/CSV spreadsheet parser.
PATCH	/api/v1/admin/students/:id/status	Admin	Toggle student academic status between Active and Inactive/Suspended.
GET	/api/v1/admin/faculty	Admin	Retrieve registered academic faculty members with assigned courses.
POST	/api/v1/admin/faculty	Admin	Add a new professor and assign designated teaching

			modules.
POST	/api/v1/admin/faculty/bulk-import	Admin	Bulk import faculty staff from Excel/CSV spreadsheets.
DELETE	/api/v1/admin/faculty/:id	Admin	Remove faculty member from the institutional roster.
GET	/api/v1/admin/departments	Admin, Faculty, Student	Retrieve academic department list, department heads, and course catalogs.
POST	/api/v1/admin/departments	Admin	Create new academic department with code, head, and initial syllabus.
PUT	/api/v1/admin/departments/:id	Admin	Update department details, student capacity, and department head designation.
DELETE	/api/v1/admin/departments/:id	Admin	Remove department structure from the academic catalog.
POST	/api/v1/admin/departments/:id/courses	Admin	Add a new accredited course module with credits and semester.
DELETE	/api/v1/admin/departments/:id/courses/:courseCode	Admin	Remove a course module from a department syllabus.
GET	/api/v1/admin/settings	Admin	Fetch global system configurations like attendance thresholds and maintenance mode.
PUT	/api/v1/admin/settings	Admin	Update global platform settings and system operational thresholds.
GET	/api/v1/admin/ai-metrics	Admin	Retrieve AI platform metrics including token usage and query volume.

GET	/api/v1/admin/search	Authenticated	Global search across courses, drives, clubs, and students.
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3. Faculty Operations & Academics (/api/v1/faculty)

Method	Endpoint	Access	Description
GET	/api/v1/faculty/dashboard	Faculty, Admin	Fetch faculty dashboard summary with enrolled students and pending reviews.
GET	/api/v1/faculty/students	Faculty, Admin	Retrieve student roster for attendance marking and grading.
POST	/api/v1/faculty/attendance	Faculty, Admin	Submit and record a live classroom attendance session in MongoDB.
GET	/api/v1/faculty/attendance/session	Faculty, Admin	Fetch attendance session records for a specific date, subject, and time slot.
GET	/api/v1/faculty/assignments	Faculty, Admin	Retrieve coursework modules created and published by faculty.
POST	/api/v1/faculty/assignments	Faculty, Admin	Create and publish a new student assignment with due dates and rubrics.
DELETE	/api/v1/faculty/assignments/:id	Faculty, Admin	Delete an assignment and cascade remove student submissions.
GET	/api/v1/faculty/submissions	Faculty, Admin	Fetch student assignment submissions roster across all assigned subjects.
PUT	/api/v1/faculty/submis	Faculty, Admin	Grade student assignment

	sions/:id/grade		submission with marks and feedback.
POST	/api/v1/faculty/notices	Faculty, Coordinator, Admin	Publish campus announcements, exam circulars, and study materials.
GET	/api/v1/faculty/notices	Authenticated	Retrieve all published campus announcements, circulars, and notices.
DELETE	/api/v1/faculty/notices/:id	Faculty, Coordinator, Admin	Remove published circular or study material notice.
POST	/api/v1/faculty/courses	Faculty, Admin	Add assigned subject to faculty teaching catalog.
DELETE	/api/v1/faculty/courses/:code	Faculty, Admin	Remove assigned subject from teaching roster.
PUT	/api/v1/faculty/courses/:code/syllabus	Faculty, Admin	Update course syllabus, lecture modules, and reference books.

4. Student & Extracurricular Coordinator Operations (/api/v1/coordinator)

Method	Endpoint	Access	Description
GET	/api/v1/coordinator/stats	Coordinator, Admin	Fetch real-time coordinator metrics like events and digital passes.
POST	/api/v1/coordinator/approvals/:id/decide	Coordinator, Admin	Approve or decline student club membership or event organizer pass requests.

5. Attendance Management Engine (/api/v1/attendance)

Method	Endpoint	Access	Description
GET	/api/v1/attendance/student	Student, Authenticated	Retrieve student aggregate attendance %, monthly trends, and subject breakdowns.
GET	/api/v1/attendance/summary	Student, Authenticated	Fetch condensed attendance summary and 75% university eligibility status.
POST	/api/v1/attendance/sessions	Faculty, Admin	Create and persist a classroom attendance session.

6. Student Coursework & Submissions (/api/v1/assignments)

Method	Endpoint	Access	Description
GET	/api/v1/assignments	Authenticated	Retrieve all published coursework modules relevant to student department.
POST	/api/v1/assignments/submit	Student	Submit assignment solution (GitHub repo link, PDF URL, or project archive).
GET	/api/v1/assignments/my-submissions	Student	Fetch all submitted solutions with grading status, marks, and feedback.

7. Campus Events & Digital QR Passes (/api/v1/events)

Method	Endpoint	Access	Description
GET	/api/v1/events	Authenticated	List all upcoming campus hackathons,

			symposiums, and cultural events.
POST	/api/v1/events	Coordinator, Admin	Publish a new campus event with venue, date, organizer, and registration links.
DELETE	/api/v1/events/:id	Coordinator, Admin	Delete an event and revoke associated registration passes.
POST	/api/v1/events/:id/register	Student	Register for event and generate digital QR entry pass with encrypted token.
GET	/api/v1/events/my-registrations	Student	Retrieve student registered event passes and dynamic QR codes.
DELETE	/api/v1/events/registrations/:id	Student	Cancel event registration and release reserved ticket pass.

8. Placement & Career Management Hub (/api/v1/placements)

Method	Endpoint	Access	Description
GET	/api/v1/placements/drives	Authenticated	Retrieve active corporate recruitment drives, CTC packages, and eligibility.
POST	/api/v1/placements/drives	Admin	Post a new corporate placement drive with company, package, and criteria.
DELETE	/api/v1/placements/drives/:id	Admin	Delete recruitment drive and archive student applications.
POST	/api/v1/placements/:id/check-eligibility	Student	AI-assisted eligibility verification against CGPA, backlogs, and

			skills.
POST	/api/v1/placements/:id/apply	Student	Submit job application with verified student profile and resume PDF.
GET	/api/v1/placements/my-applications	Student	Track submitted placement applications and interview status.

9. Clubs & Student Societies (/api/v1/clubs)

Method	Endpoint	Access	Description
GET	/api/v1/clubs	Authenticated	Retrieve registered technical, cultural, and social student clubs.
POST	/api/v1/clubs/:id/join	Student	Join a student society and increment membership roster.
DELETE	/api/v1/clubs/:id/leave	Student	Leave a student society and update active memberships.

10. Grievance & Complaint Redressal (/api/v1/complaints)

Method	Endpoint	Access	Description
GET	/api/v1/complaints	Authenticated	Fetch submitted grievance tickets with resolution status.
POST	/api/v1/complaints	Authenticated	File an anonymous or verified campus grievance ticket.

11. Real-time Notifications Engine (/api/v1/notifications)

Method	Endpoint	Access	Description
GET	/api/v1/notifications	Authenticated	Fetch real-time in-app notification alerts with unread counter.
PATCH	/api/v1/notifications/:id/read	Authenticated	Mark specific notification alert as read.

12. CampusGPT AI Copilot & Chat Engine (/api/v1/ai)

Method	Endpoint	Access	Description
POST	/api/v1/ai/query	Authenticated	Multi-tenant AI workspace query with role-tailored prompt engine.
GET	/api/v1/ai/history	Authenticated	Retrieve saved conversation thread history and past message trajectories.
DELETE	/api/v1/ai/history/:id	Authenticated	Delete conversation thread from chat history.

13. System Health & Monitoring

Method	Endpoint	Access	Description
GET	/api/v1/health	Public	Health check endpoint returning API operational status and timestamp.